Subject Code: FYP4112

Programme Code: CT206

EDIT COLOR - BLACK!

PROJECT BRIEF FORM

To be submitted BY: WEEK 4

1. Your details:

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2. Project title (provisional) [Meaningful, relevant and concise]

CyberSafe Learning Hub

3. Intended user or group of users and client. [a) Who is the intended user or group of users and client? b) Why do you think there is a need for this project?

3a. Intended user and client:

- User
  - The intended users of this system are the university students of Kolej Teknologi Antarabangsa Cybernetics. They represent a group of young digital natives who are active online but often lack sufficient cybersecurity knowledge and awareness. By engaging them through interactive lessons and gamified challenges, the system aims to improve their understanding of safe online practices and reduce their vulnerability to cyber threats.

- Client
  - The client for this project is Kolej Teknologi Antarabangsa Cybernetics (KTAC). As a higher education institution responsible for preparing students to face an increasingly digital society, KTAC plays a crucial role in shaping digital literacy. By adopting this system, the university can integrate cybersecurity awareness into its educational framework, allowing lecturers to monitor student progress and provide structured, engaging training materials that strengthen safe online behaviors.

3b. Problem Statement:

1. Low Cybersecurity Awareness Among Students

   University and college students both generally and specifically in Malaysia often lack sufficient understanding of cybersecurity threats such as phishing, malware, and unsafe online behavior, leaving them vulnerable to exploitation. For instance, a study of Malaysian secondary and tertiary students found that although willingness to learn about cybersecurity is high, their actual understanding of such threats remains limited, highlighting a pressing need for structured awareness education (Zulkifli et al.,

2020).

2. Limited Access to Interactive Cybersecurity Education
   Traditional school-based awareness methods (such as lectures and pamphlets) are often boring or difficult for teenagers to understand, leading to weak engagement and low knowledge retention. Research shows that passive teaching methods result in limited behavioral change, while interactive and experiential approaches are more effective in improving cybersecurity awareness among youth (Ng & Kamsin, 2021).

3. Lack of Motivation to Practice Cybersecurity Skills
   Without gamified or hands-on experiences, students are less likely to apply cybersecurity knowledge in real life, leaving them vulnerable to online threats. Studies highlight that gamification techniques, such as quizzes, badges, and leaderboards, can significantly boost motivation and engagement in cybersecurity education (CM Alliance, 2025).

4. Objectives and project deliverables. [a) What are the characteristics/properties that the final product should possess? b) What are the process stages and the corresponding deliverables that will enable you to create the final product?]

4a. Objective:

1. To Improve Cybersecurity Awareness Among Students
   This objective focuses on raising students' understanding of fundamental cybersecurity threats such as phishing, malware, and weak passwords. By introducing structured, age-appropriate modules, students will learn not only the definitions of these threats but also practical steps they can take to stay safe online. Achieving this objective will help bridge the gap between students' existing knowledge and the level of awareness needed to protect themselves effectively in the digital environment.

2. To Provide Interactive and Engaging Cybersecurity Education
   This objective emphasizes the need to move beyond traditional lecture-style teaching, which is often less effective in helping students retain knowledge. The platform will include interactive elements such as short quizzes, multiple-choice questions, and visual content such as infographics to make learning more engaging. By making lessons easy to follow and interactive, students are more likely to stay attentive, understand key concepts, and apply them in their digital lives.

3. To Motivate Students to Apply Cybersecurity Skills in Practice
   This objective aims to encourage students to remain motivated while learning cybersecurity by introducing gamification features. The platform will reward students with points for completing lessons and quizzes, display their learning progress, and award badges when milestones are achieved. These simple elements of gamification help make learning feel rewarding and fun, which increases the likelihood that students will apply what they learn in their daily online activities.

4b. Agile Methodology

Agile methodology is chosen because it allows the project to be developed in **small, manageable iterations (sprints)** with continuous testing and feedback from both students and lecturers. This flexibility ensures that any new requirements or usability improvements can be quickly adapted without delaying the whole project. Agile also reduces risks by allowing features such as authentication, lessons, and gamification to be built and refined step by step, ensuring the final product is both functional and engaging.

Phase	Description	Deliverables
Plan	Planning is crucial in application development. At this stage, user requirements, client requirements, and project methods are identified. Research is conducted on cybersecurity awareness gaps and how gamification can motivate students.	Proposal document, requirement analysis, method selection (Laravel, MySQL, Google OAuth, Bootstrap).
Design	The design phase focuses on creating enough structure to guide development. This includes preparing the flowchart of processes, wireframes of dashboards, UI design templates, and the database design for modules, lessons, challenges, progress, and badges.	Flowchart diagrams, wireframes, UI mockups, database schema design.
Develop	This phase involves incremental coding, integration, and testing of features. Core functions such as authentication, role-based dashboards, CRUD for modules/lessons/challenges, and gamification (points, badges, progress tracking) are built.	Prototype of the CyberSafe Learning Hub application.
Test	Testing is iterative and continuous. Unit tests, integration tests, and system tests are conducted alongside development. A pilot test with students and teachers is performed to validate usability and effectiveness.	Unit testing reports, integration testing reports, system testing results, user acceptance testing feedback.
Deploy	The platform is deployed on a staging server for trials and later in a production environment accessible to schools and students. Deployment includes pipeline setup, security configurations, and ensuring system accessibility.	Fully deployed CyberSafe Learning Hub application.
Review	After deployment, the system is reviewed based on teacher and student feedback. Maintenance is prioritized to fix bugs and optimize performance. Feedback is collected via surveys (Google Forms).	Google Form survey results, post-deployment evaluation report, updated features.

5. Research [a) How will you investigate/identify in detail the needs of the specified user in (3) b) How will you investigate the background of the project?]

5a. To investigate the needs of the intended users, a questionnaire will be distributed to at least 50 students. The purpose of the questionnaire is to evaluate their current level of cybersecurity awareness, identify their preferred learning methods, and measure their interest in using interactive or gamified educational tools.

In addition, an interview will be conducted with the lecturer in charge to gather insights from the educator's perspective. This will help identify the challenges faced in delivering cybersecurity awareness, ensure that the platform aligns with the college's requirements, and validate the relevance of the proposed system in supporting teaching and learning activities.

5b. Literature Review

Three existing cybersecurity awareness and training platforms that can be used as references for this project are TryHackMe, Cybrary, and HackTheBox Academy.

- **TryHackMe**

  TryHackMe is an online training platform that provides virtual labs and challenges to teach cybersecurity concepts ranging from basic networking to advanced penetration testing. The platform emphasizes gamification through badges, levels, and interactive tasks, making it engaging for learners. However, its content is often technical and more suitable for university students or professionals, rather than high school students or non-IT university or college students. This highlights the need for a simpler, more beginner-friendly system tailored for entry-level learners.

- **Cybrary**

  Cybrary is a cybersecurity education platform that delivers structured courses, hands-on labs, and certification training. It is widely used by IT professionals to upskill in specialized domains like incident response, network defense, and cloud security. While it offers a broad library of resources, it is less suited for teenagers or non-IT university students because of its professional focus and complexity. Nevertheless, Cybrary demonstrates how structured modules and clear learning paths can help organize content in a way that promotes progression.

- **HackTheBox Academy**

  HackTheBox Academy is a practical training environment designed to teach ethical hacking and penetration testing skills through real-world labs and simulations. It provides deep, hands-on experience with cybersecurity tools and attack methods. While effective for advanced learners, the platform's technical depth makes it inaccessible to beginners without prior knowledge. Despite this limitation, HackTheBox Academy illustrates the value of hands-on training in cybersecurity education, which can inspire simplified interactive elements for younger or entry-level learners.

6. Evaluation. [a) What will make your product successful? b) How will you demonstrate that your product fulfils the needs of the user in (3)?

6a. To ensure the CyberSafe Learning Hub truly meets user needs and is a strong tool against digital risks, we will adopt a thorough testing strategy. This will include Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT). Each type of testing will target different aspects of the system to guarantee functionality, reliability, and usability.

- **Unit Testing**
  - **Objective:** Verify that all individual components or units of code work as intended.
  - **Steps:**
    • Identify units for testing: focus on the smallest parts of the application, such as login validation, quiz scoring, and badge assignment.
    • Automate tests using Laravel's PHPUnit framework to ensure smooth execution without errors or bugs.
  - **Examples:**
    • Test the login function to ensure it validates correct and incorrect credentials.
    • Test the quiz scoring function to ensure points are correctly awarded.
- **Integration Testing**
  - **Objective:** Ensure that all modules or services in the application work correctly together.
  - **Steps:**
    • Simulate processes where multiple components interact. For example, completing a lesson should update progress and award points automatically.
    • Test integration between user progress tracking and the badge system.
  - **Examples:**
    • Ensure completing a quiz updates the progress table and awards points.
    • Verify that reaching a milestone triggers badge assignment.
- **System Testing**
  - **Objective:** Conduct a comprehensive end-to-end assessment of the application to ensure it meets requirements.
  - **Steps:**
    • Perform tests that simulate real-world usage scenarios.
    • Ensure dashboards, lessons, challenges, and gamification features work seamlessly together.
  - **Examples:**
    • Verify that students can log in, complete a lesson, take a quiz, and see their progress update correctly.
    • Confirm that admins can create/edit modules and monitor student performance.

6b. We will conduct User Acceptance Testing (UAT) with high school students and teachers (clients). Success will be defined using explicit acceptance criteria, such as correct lesson completion tracking, accurate quiz scoring, and badge assignment.

- **User Acceptance Testing (UAT)**
  - **Objective:** Validate the application against user requirements and confirm it meets the needs of the end users.
  - **Steps:**
    - Conduct UAT sessions with selected college students and lecturers.
    - Collect structured feedback on usability, engagement, and effectiveness.
    - Conduct interviews with both students and lecturers to validate alignment with real classroom needs.
    - Perform both functional and non-functional testing to ensure the system meets expectations in real-world use.
  - **Functional Testing Examples:**
    - Students test lessons and quizzes to verify that scores are calculated correctly.
    - Lecturers test the admin dashboard to ensure they can create, edit, and track modules successfully.
    - Authentication is tested to confirm both manual and Google login function correctly.
    - Gamification features (points, badges, progress tracking) are tested to validate accurate rewards and updates.
  - **Non-Functional Testing Examples:**
    - **Usability:** Students provide feedback on ease of navigation, clarity of instructions, and engagement level.
    - **Performance:** Test response time when loading lessons, quizzes, and dashboards under typical usage.
    - **Security:** Validate that sensitive data (e.g., student accounts, progress records) is protected through secure login and encryption.

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7.Development skills. [a) What information and resources (software and hardware) do you need to complete the project? b) Which of these do you need to acquire yourself?]

7a. **Information and resources needed to complete the project:** HARDWARE??

To complete the CyberSafe Learning Hub project, the following information and resources are required:

- **Laravel 12 framework** for backend development.
- **MySQL database** for storing user accounts, modules, lessons, challenges, and progress.
- **Bootstrap (or TailwindCSS)** for responsive frontend design.
- **Google OAuth API** for Google login integration.
- **PHPUnit (Laravel testing framework)** for automated testing.
- **Cybersecurity awareness content** (phishing, password safety, safe browsing, etc.) gathered from trusted sources such as NCSA, OWASP, or academic journals.

7b. Some resources are readily available, while others require additional effort to acquire:

- To Acquire Ourselves:
  - Mastery of advanced Laravel features (authentication, middleware, role-based access).
  - Implementation of gamification elements (points, badges, progress tracking).
  - Knowledge of web application security practices (e.g., encryption, secure session handling, signed URLs).
  - Research and adaptation of cybersecurity awareness content.
- Already Available / Accessible:
  - Laravel framework and MySQL (open source).
  - Bootstrap/Tailwind (open source).
  - Google OAuth API (free integration).
  - PHPUnit testing tools (included in Laravel).

8. Skill acquisition. [How do you intend to gain the skills, information and resources specified in (7)?]

To successfully build the CyberSafe Learning Hub, we will acquire the necessary skills through **self-learning, practical training, and online research**. These approaches will provide both technical expertise and relevant content knowledge to ensure the system is functional, engaging, and secure.

1) Self-learning (Technical Skills)

We will dedicate time to building the technical foundation of the project through self-learning. This includes gaining proficiency in Laravel for backend development, MySQL for database management, Bootstrap/Tailwind for designing user-friendly interfaces, and Google OAuth API for secure third-party login. By following structured tutorials and official documentation, we can gradually develop confidence in building core functionalities such as authentication, CRUD operations, and gamification features. The goal is to ensure that every component of the system is robust and well-implemented.

Resources:

- Laravel Documentation: https://laravel.com/docs
- FreeCodeCamp.org (2021). *Laravel Tutorial for Beginners* (YouTube): https://www.youtube.com/watch?v=MFh0Fd7BsjE
- W3Schools. *MySQL Tutorial*: https://www.w3schools.com/mysql/
- Bootstrap Documentation: https://getbootstrap.com/docs/
- TailwindCSS Documentation: https://tailwindcss.com/docs
- Google Developers. *Using OAuth 2.0*: https://developers.google.com/identity/protocols/oauth2

2) Training (Practical Application)

Beyond theory, we will engage in hands-on training by building small prototypes and testing individual modules before integrating them into the full system. This includes experimenting with login systems, testing role-based access, creating sample quizzes, and implementing gamification elements such as points and badges. We will also practice automated testing using PHPUnit to validate the accuracy and reliability of our code. This stage will bridge the gap between learning and application, ensuring that the skills we acquire are directly tested in a development environment.

Resources:

- Traversy Media (2020). *Laravel From Scratch* (YouTube): https://www.youtube.com/watch?v=ImtZ5yENzgE
- Laravel Docs – Testing: https://laravel.com/docs/testing

3) Research (Content and Best Practices)

In addition to technical learning, we will conduct **online research** to ensure that the content and security practices embedded in the project are accurate, reliable, and aligned with global standards. Research will cover three key areas which was cybersecurity awareness content as example phishing, password safety, safe browsing. Then, secure coding practices to ensure the platform is safe from common vulnerabilities and lastly, gamification studies to validate the motivational strategies used in the platform. These insights will ensure that the CyberSafe Learning Hub is not only technically functional but also pedagogically effective and aligned with real-world cybersecurity practices.

Resources:

- OWASP Foundation: https://owasp.org
- National Cyber Security Alliance (NCSA). *Annual Cybersecurity Attitudes and Behaviors Report*: https://staysafeonline.org
- Bhavsar, V., Kadlak, A., & Sharma, S. (2018). *Study on Phishing Attacks*. International Journal of Computer Applications. DOI: https://doi.org/10.5120/ijca2018918286
- Nwokeji, J., Matovu, M., & Rawal, R. (2020). The use of gamification to teach cybersecurity awareness in information systems. *Proceedings of the SIGED: IAIM Conference 2020*, 29, 1–7. https://aisel.aisnet.org/siged2020/29

9. Outline (synopsis) of your project. [What are the aims and objectives of the project?]

This project aims to develop a web-based cybersecurity awareness platform for college students, called CyberSafe Learning Hub. The system is designed to deliver interactive lessons and quizzes on topics such as phishing, password safety, and safe browsing practices. To ensure the learning process remains engaging and motivating, the platform integrates gamification features such as points, badges, and progress tracking. Lecturers will act as administrators, with the ability to create and manage content, as well as monitor student progress through an administrative dashboard.

Cybersecurity awareness among college students remains limited, particularly for those outside of IT-related programs. Many students frequently use digital devices and online platforms but lack sufficient knowledge to protect themselves against common threats such as phishing, malware, and unsafe online behaviors. Research has shown that while students may have some awareness of cyber risks, their actual practices and protective actions remain weak, revealing a clear gap between knowledge and application (Zulkifli et al., 2020). Similarly, Ng and Kamsin (2021) emphasized that traditional lecture-based methods are not effective in maintaining student engagement or ensuring long-term knowledge retention. Gamification, on the other hand, has been proven to significantly improve learner motivation and outcomes in cybersecurity education (Wu et al., 2021; Gwenhure & Rahayu, 2024).

The aim of this project is to improve cybersecurity awareness among students by providing structured, beginner-friendly lessons that cover essential concepts and practical strategies for staying safe online. It also seeks to make learning more engaging by moving away from static lecture materials and incorporating

quizzes, interactive content, and visual resources. Additionally, the project intends to motivate students to consistently apply cybersecurity practices in their daily lives by rewarding them with gamification features such as points, badges, and visible progress tracking.

Through these objectives, the CyberSafe Learning Hub will bridge the gap between advanced, professional-level cybersecurity training platforms and the absence of accessible, student-oriented tools for non-IT learners. By focusing on accessibility, interactivity, and motivation, the project will empower students at Kolej Teknologi Antarabangsa Cybernetics to adopt safer online behaviors while also providing lecturers with a sustainable platform to enhance digital literacy and cybersecurity education.

10. Estimate the number of hours you are planning to spend for each of the following tasks:

Background research and learning new skills	70
Requirements gathering and investigation	30
Product design	40
Product development	100
Product evaluation	25
Final report preparation	25
Other (Presentation Rehearsal)	10
Total number of hours	**300**

Signature..... Date: 28/8/2025